

Assessing Harmful Algal Blooms in Mississinewa Lake: a Current Progress Report with 2025 Bloom Conditions

Isaac Bradford¹, Nileshwari Yewle¹, Nhu Hoang Van Pham², Sheng Tan², Cary Troy², Melba Crawford², Zhi Zhou², Keith Cherkauer¹

¹Department of Agricultural and Biological Engineering, Purdue Univ., West Lafayette, IN 47907;

²Lyles School of Civil and Construction Engineering, Purdue University



Agricultural and Biological Engineering

OVERVIEW

- Monitoring and evaluating water quality is crucial for understanding and addressing environmental and human health issues.
- As water quality stress increases, Harmful Algal Blooms (HABs) continue to propagate and cause problems.
- HABs can have a negative impact on ecosystems by contributing to eutrophication, and releasing harmful toxins.
- Goal is to generate a model using remote sensing to predict the occurrence of HABs in inland waters of the North-Central U.S.



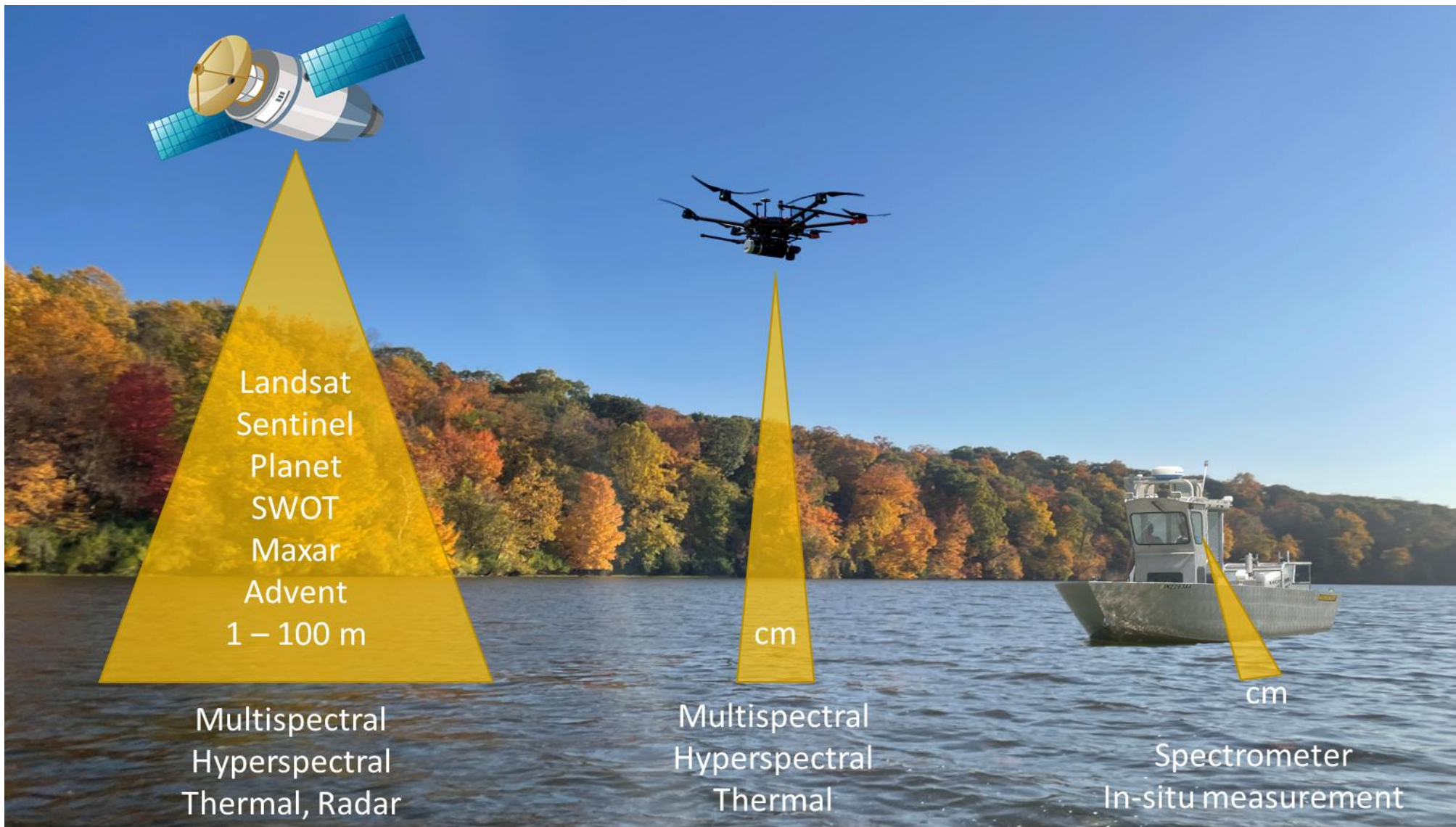
Left: Indiana with points at all 4 sampled Lakes: from left to right Shafer, Freeman, , Mississinewa, Salamonie)

OBJECTIVES

1. Rapidly quantify the biogeochemical state of inland waters in the North-Central U.S. by analyzing their optical and thermal properties via remote sensing
2. To identify preconditions that lead to harmful algal blooms (HABs) and predict the potential occurrence of these blooms in these water bodies.

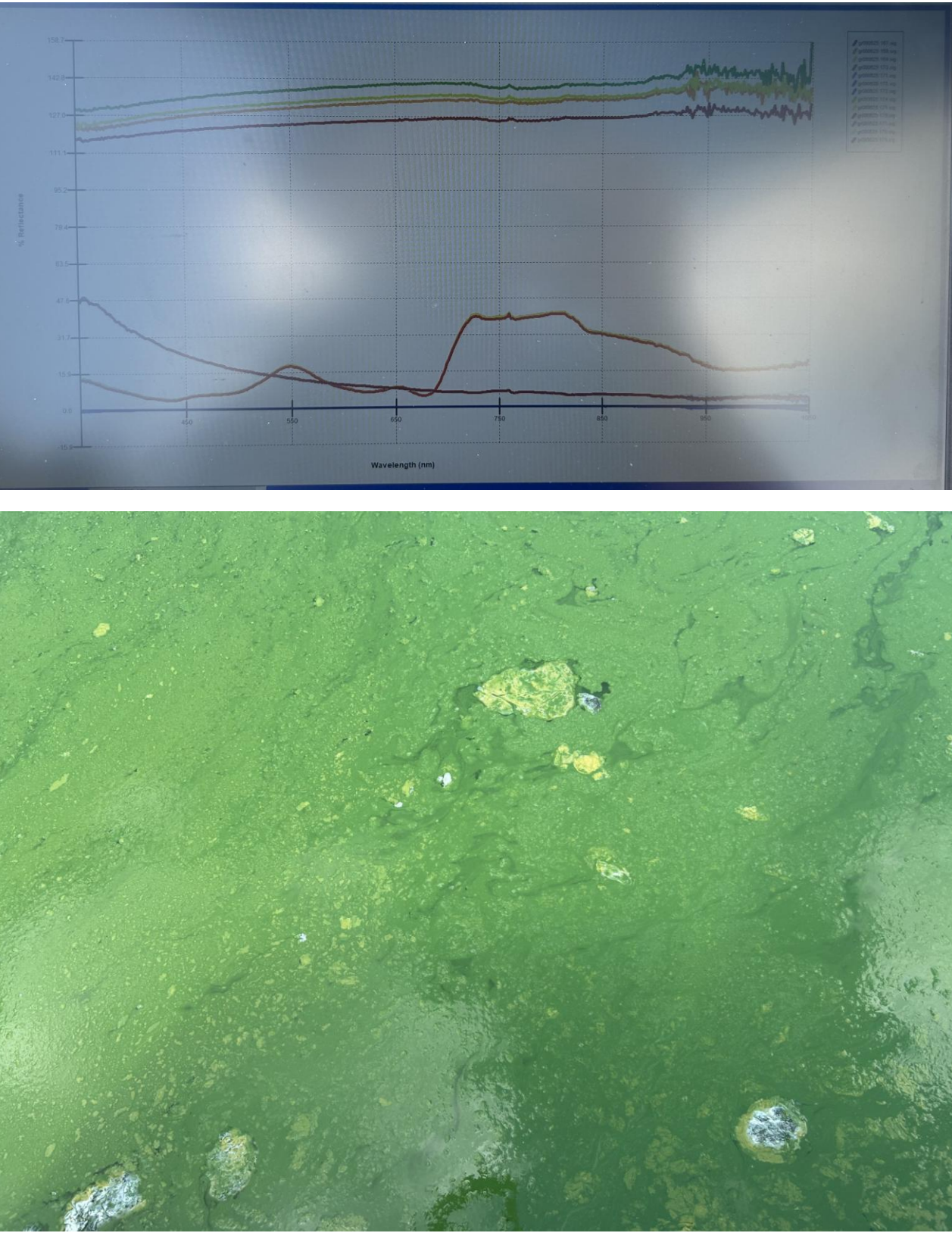
SAMPLING METHODOLOGY

- In-situ **spectral reflectance** from water's surface recorded with GER1500
- **Water temperature, ODO, conductivity or specific conductance, oxidation reduction potential, TDS or turbidity and pH** vertical profiles with YSI multiparameter sonde
- Water samples collected undergo lab analysis
 - **Microcystin** with U.S. EPA Method 546
 - **Chlorophyll-a** with U.S. EPA Method 445.0
 - **Soluble nutrients** ammonia, nitrate-nitrite, and orthophosphate, and total phosphorus analyzed with U.S. EPA Methods 103, 114, 118365.1
- Imagery from remote sensing platforms collected and evaluated post in-situ sampling.



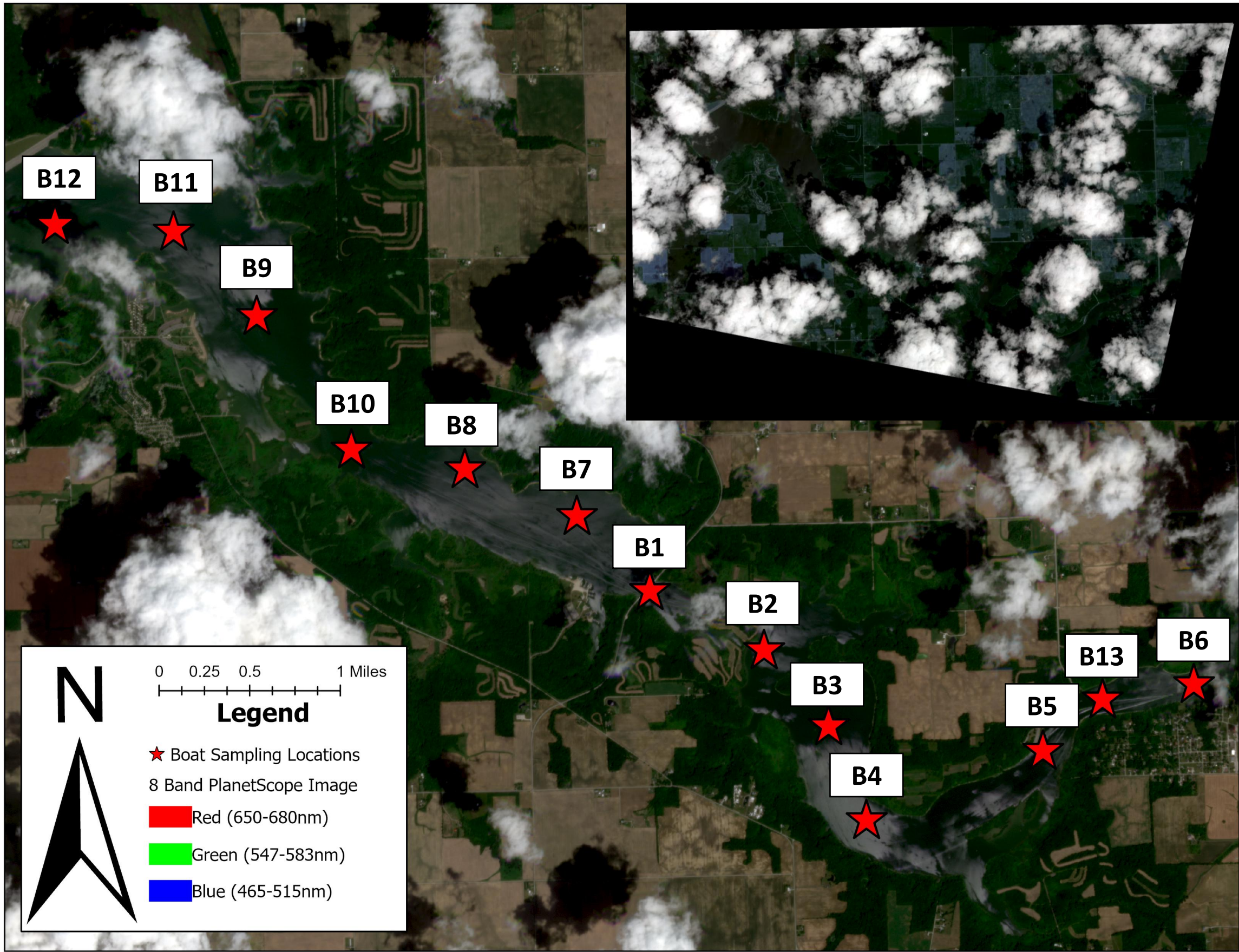
Data collection platforms and scale of sampling

FIELD COLLECTION



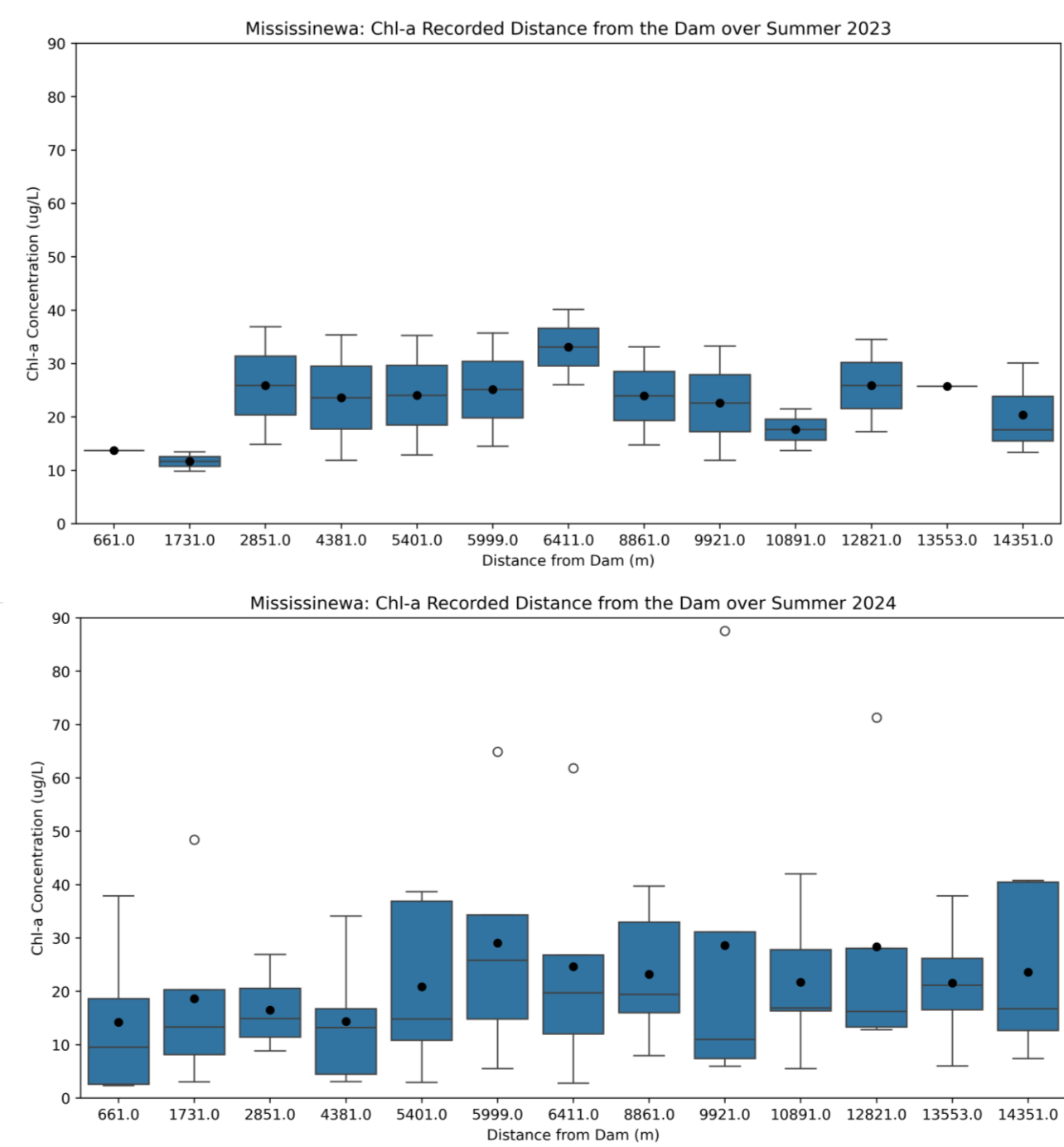
Top: What we see on the boat for our ground truth collection (GER1500 handheld spectrometer) Used for remote sensing calibration

Bottom: The worst algae we saw on Mississinewa this year 8/6/2025



Eight-band SuperDove PlanetScope Image for Lake Mississinewa 6/24/2024. Inset shows cloud cover from 7/18/24 for comparison.

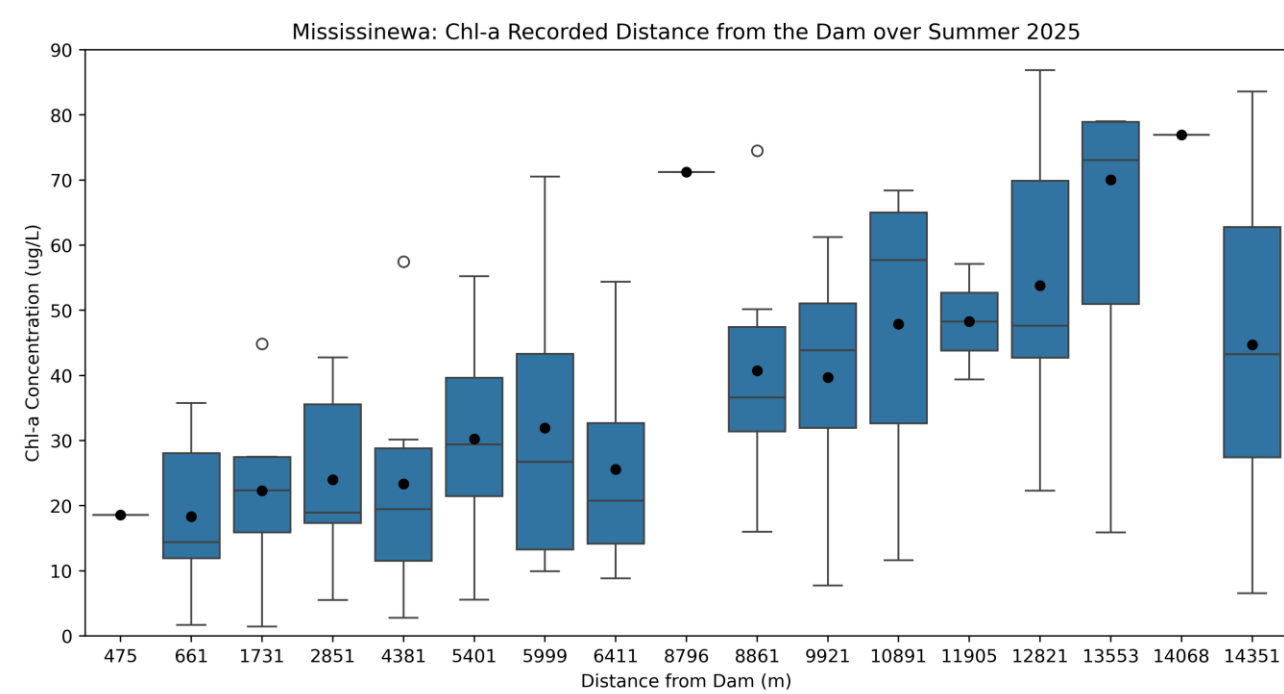
RESULTS



These graphs top to bottom are boxplots of Chlorophyll-a concentrations by location for each year (2023-2025) over the sampling season (May-Sep)

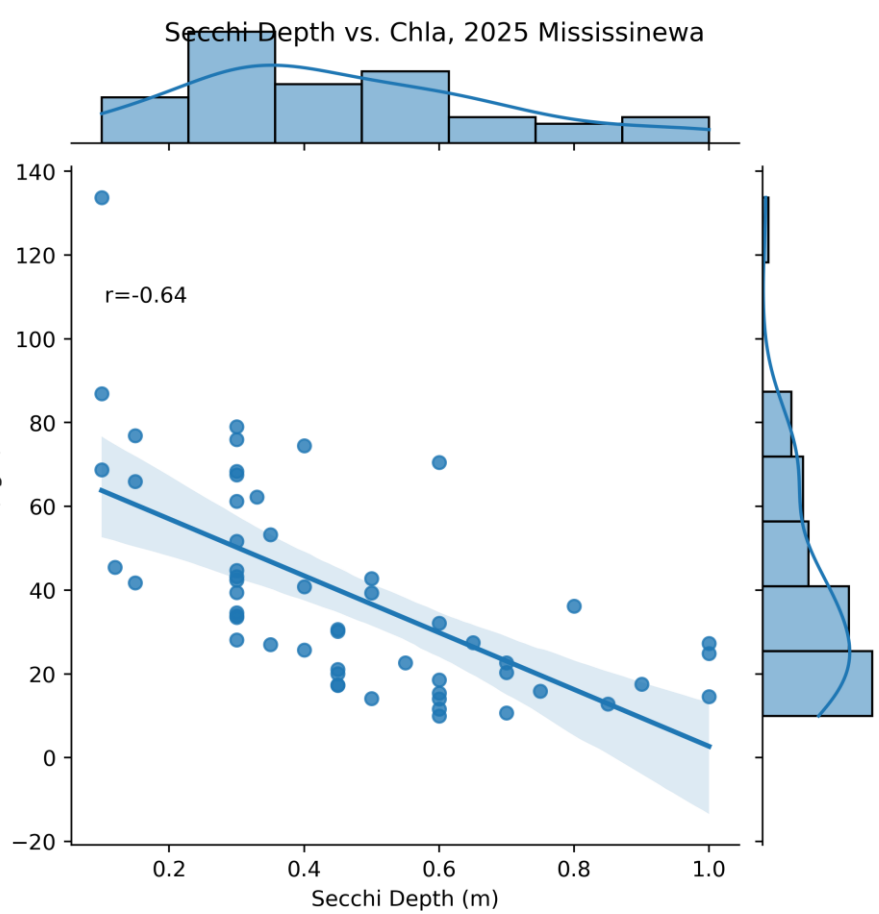
2025 had both the largest range of data and the highest average concentration

This was due to the bloom conditions (see bottom left photo) that had not seen similar results since 2019



Right: 2025 Secchi Disk Depth vs, Chl-a Concentrations, finding a negative relationship with Chla. Turbidity as a separate metric to be added later

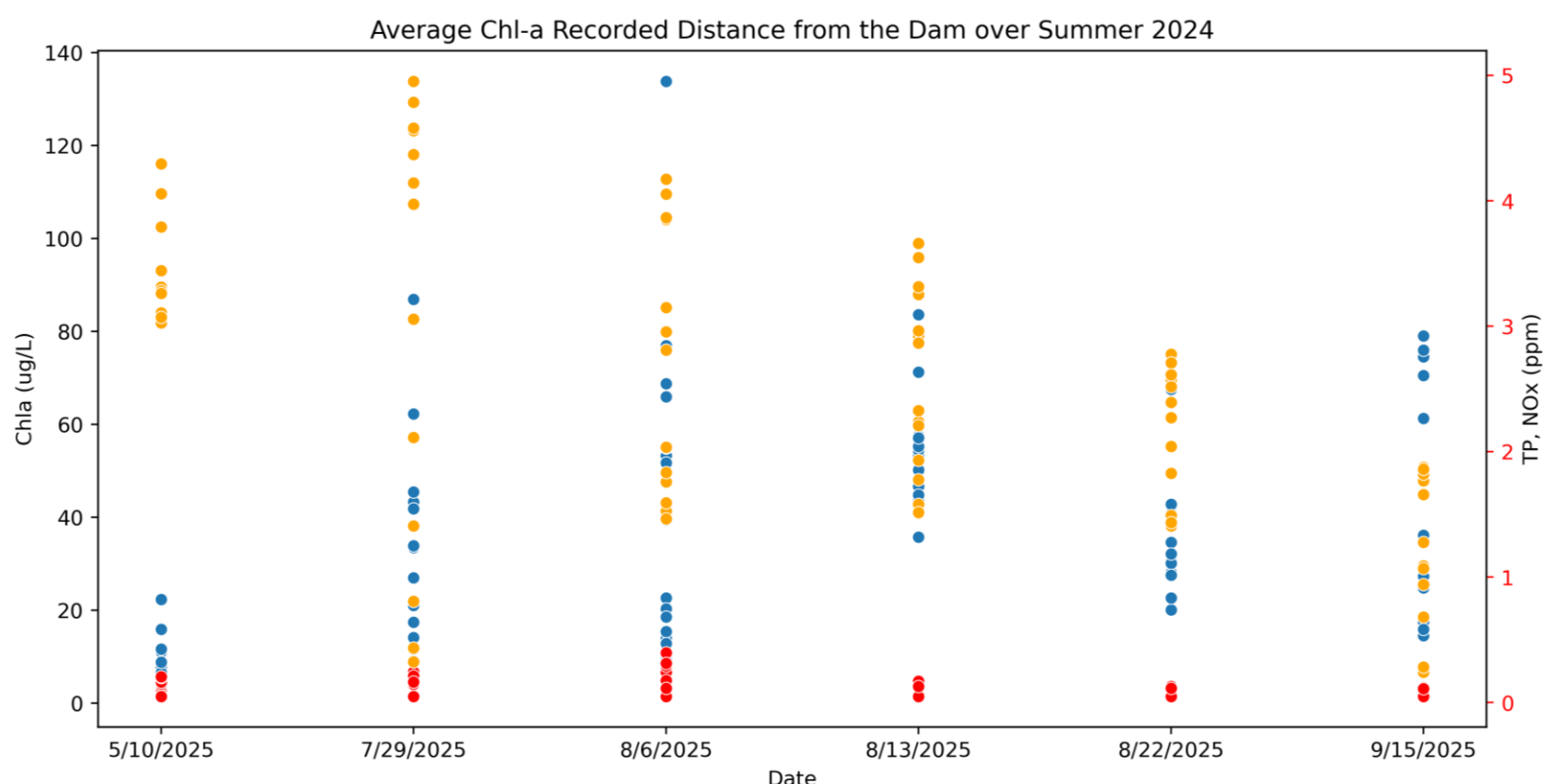
We are currently analyzing the differences in conditions that led to the blooms occurring 2025 but not the other years



Thank you to everyone who has helped us on this project and assisted with us with getting our data!



CONCLUSIONS



2025 Chla, Nitrate-Nitrite, and Total Phosphorous by Date (points from all locations included)

- There seemed to be an increase in TP and NOx while there was a higher reading of Chla
- There needs to be more research done on the temperature and the pre-bloom conditions that led to such high nutrient and intense blooming
- The Chla recored this year was the **highest of all 3 years** and Mississinewa was the most intense bloom in **all of Indiana in 2025**

FUTURE WORK

- Utilize arcpy within ArcGIS to automate the remotely sensed spectral extraction
- Analyze biogeochemical data paired with remotely sensed parameters to determine patterns
- Present updates and findings from lab analysis to community.
- Develop a predictive machine learning model for HAB risk using these data with supplementary information on land use and weather.

Special thanks to Mandy Limiac, Areesha Saleem and Max Ahrbecker.

Figure 5, Burket, M., Olmanson, L., & Brezonik, P. (2023). Comparison of Two Water Color Algorithms: Implications for the Remote Sensing of Water Bodies with Moderate to High CDOM or Chlorophyll Levels. *Sensors*, 23, 1071. <https://doi.org/10.3390/s23031071>

This research is funded by NASA (NNH21ZDA001N-RSWQ).